

Machine Learning-Lecture 3

P1.py

```
C:\ml\L3>python p1.py
```

```
area bedrooms price
```

```
0 2.6      3 5.50
```

```
1 3.0      3 5.65
```

```
2 3.2      3 6.10
```

```
3 3.6      3 6.95
```

```
4 4.0      4 7.60
```

```
enter area 28
```

```
enter number of bedrooms 4
```

```
[43.13846154]
```

P2.py

```
C:\ml\L3>python p2.py
```

```
area bedrooms price
```

```
0 2.6      3 5.50
```

```
1 3.0      3 5.65
```

```
2 3.2      3 6.10
```

```
3 3.6      3 6.95
```

```
4 4.0      4 7.60
```

```
enter area 25
```

```
enter number of bedrooms 3
```

```
38.478846153846156
```

P3.py

```
C:\ml\L3>python p3.py
```

```
area bedrooms price
```

```
0 2.6      3 5.50
```

```
1 3.0      3 5.65
```

```
2 3.2      3 6.10
```

```
3 3.6      3 6.95
```

```
4 4.0      4 7.60
```

```
b0 = 0.8076923076923093
```

```
b1 = 1.480769230769231
```

```
b2 = 0.2173076923076914
```

P4.py

```
C:\ml\L3>python p4.py
```

```
model creation done
```

P5.py

```
C:\ml\L3>python p5.py
```

```
enter area 25
```

```
enter number of bedrooms 3
```

```
[38.47884615]
```

P6.py

```
C:\ml\L3>python p6.py
```

```
R&D Spend Administration Marketing Spend Profit
```

```
0 165349.20 136897.80 471784.10 192261.83
```

```
1 162597.70 151377.59 443898.53 191792.06
```

```
2 153441.51 101145.55 407934.54 191050.39
```

```
3 144372.41 118671.85 383199.62 182901.99
```

```
4 142107.34 91391.77 366168.42 166187.94
```

```
5 131876.90 99814.71 362861.36 156991.12
```

```
6 134615.46 147198.87 127716.82 156122.51
```

7	130298.13	145530.06	323876.68	155752.60
8	120542.52	148718.95	311613.29	152211.77
9	123334.88	108679.17	304981.62	149759.96
10	101913.08	110594.11	229160.95	146121.95
11	100671.96	91790.61	249744.55	144259.40
12	93863.75	127320.38	249839.44	141585.52
13	91992.39	135495.07	252664.93	134307.35
14	119943.24	156547.42	256512.92	132602.65
15	114523.61	122616.84	261776.23	129917.04
16	78013.11	121597.55	264346.06	126992.93
17	94657.16	145077.58	282574.31	125370.37
18	91749.16	114175.79	294919.57	124266.90
19	86419.70	153514.11	0.00	122776.86
20	76253.86	113867.30	298664.47	118474.03
21	78389.47	153773.43	299737.29	111313.02
22	73994.56	122782.75	303319.26	110352.25
23	67532.53	105751.03	304768.73	108733.99
24	77044.01	99281.34	140574.81	108552.04
25	64664.71	139553.16	137962.62	107404.34
26	75328.87	144135.98	134050.07	105733.54
27	72107.60	127864.55	353183.81	105008.31
28	66051.52	182645.56	118148.20	103282.38
29	65605.48	153032.06	107138.38	101004.64
30	61994.48	115641.28	91131.24	99937.59
31	61136.38	152701.92	88218.23	97483.56
32	63408.86	129219.61	46085.25	97427.84
33	55493.95	103057.49	214634.81	96778.92
34	46426.07	157693.92	210797.67	96712.80
35	46014.02	85047.44	205517.64	96479.51
36	28663.76	127056.21	201126.82	90708.19
37	44069.95	51283.14	197029.42	89949.14
38	20229.59	65947.93	185265.10	81229.06

39	38558.51	82982.09	174999.30	81005.76
40	28754.33	118546.05	172795.67	78239.91
41	27892.92	84710.77	164470.71	77798.83
42	23640.93	96189.63	148001.11	71498.49
43	15505.73	127382.30	35534.17	69758.98
44	22177.74	154806.14	28334.72	65200.33
45	1000.23	124153.04	1903.93	64926.08
46	1315.46	115816.21	297114.46	49490.75
47	0.00	135426.92	0.00	42559.73
48	542.05	51743.15	0.00	35673.41
49	0.00	116983.80	45173.06	14681.40

	R&D Spend	Administration	Marketing Spend	Profit
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0	165349.20	136897.80	471784.10	192261.83
1	162597.70	151377.59	443898.53	191792.06
2	153441.51	101145.55	407934.54	191050.39
3	144372.41	118671.85	383199.62	182901.99
4	142107.34	91391.77	366168.42	166187.94

	R&D Spend	Administration	Marketing Spend	Profit
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45	1000.23	124153.04	1903.93	64926.08
46	1315.46	115816.21	297114.46	49490.75
47	0.00	135426.92	0.00	42559.73
48	542.05	51743.15	0.00	35673.41
49	0.00	116983.80	45173.06	14681.40

R&D Spend	0
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Administration	0
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Marketing Spend	0
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Profit	0
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dtype: int64

96.44454958404407

model saved

Business Profit Estimator

Enter R&D Spend

165349.2

Enter Admin Spend

136897.8

Enter Marketing Spend

471784.1

Estimated Profit

Profit 192586.65